

Preventing the emergence of a zoonosis following eradication of an endemic pathogen

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Abstract

Vaccination reduces the burden of human diseases and can eradicate a pathogen if it lacks an animal reservoir. The conventional view is that eradication can be followed by cessation of vaccination, but this may not be the case. This is because ceasing vaccination leads to a reduction in the level of cross-reactive immunity that suppresses related zoonoses, thereby increasing both the rate of zoonotic spillovers and human-to-human transmission following spillovers. Using mathematical models, we explore the potential for evolutionary emergence of a zoonotic pathogen following the cessation of vaccination. We demonstrate that increased human-to-human transmissibility has a profoundly greater effect on the probability of evolutionary emergence than increased spillover rate. Initial emergence with a reproductive number minimally exceeding unity enables endemic establishment, and we suggest that this establishment could facilitate further adaptation, increasing the transmissibility of the pathogen in human populations. Consequently, managing a pathogen post-emergence demands a substantially higher level of vaccination compared with what is required for proactive prevention. We discuss the implications of our framework for the emergence of mpox following cessation of smallpox vaccination.

1 Introduction

Our understanding of the use of vaccines and other intervention measures for the control and eradication of highly transmissible infectious pathogens was facilitated by the insights provided by relatively simple mathematical models of disease transmission and control [1, 30]. These models predict how the prevalence of a disease depends on its basic reproductive number R_0 , which equals the number of secondary infections arising from an infected individual introduced into a completely susceptible population [22]. Disease eradication requires vaccine (immune) coverage to exceed a fraction $1 - 1/R_0$ of the population. The successful implementation of vaccination strategies inspired by consideration of the above heuristic led to the global eradication of two diseases: smallpox in humans (1980) [3] and rinderpest in cattle (2011) and could be pertinent to other infectious diseases in the future [31]. If the eradicated pathogen lacks a zoonotic reservoir, as is the case for smallpox, the conventional view is that vaccination is no longer required and can be discontinued [5, 10, 38].

Immunity elicited by endemic human viruses (or vaccines against them) can cross-react with related zoonoses, holding them in check by preventing infection and transmission [35]. This cross-reactive immunity could reduce both the number of zoonotic spillovers as well as the transmission of the zoonoses between humans. Thus eradication followed by the cessation of vaccination could open a niche for the emergence of a related zoonoses. Indeed, after the eradication of smallpox, there has been an increase in the rate of spillovers of mpox as well as their transmission in the human population, potentially leading to the evolutionary emergence of a highly transmissible mpox disease [4, 27, 35, 36].

We explicitly consider how poorly transmissible zoonoses may accumulate transmission-increasing mutations during the stage of limited transmission after spillover, which we term "evolutionary emergence". Our goal is to develop simple models to understand the factors that govern the evolutionary emergence of a zoonosis following eradication of a related human pathogen and the subsequent waning of cross-protective immunity against the zoonosis. We consider how interventions can prevent, or at least substantially delay, the process of evolutionary emergence of the zoonosis. Smallpox is the only human pathogen which has been eradicated, following which vaccination was stopped. We illustrate the use of our framework by exploring how mpox may become more likely as population immunity to smallpox declines. In doing so we highlight knowledge gaps that limit our ability to make quantitative predictions for the evolutionary emergence of zoonoses following disease eradication.

Results and Discussion

We clarify our use of reproductive numbers. R_0 , the basic reproductive number of a pathogen equals the typical number of secondary cases arising from introduction of an individual infected into a wholly susceptible population. We let R_e equal the “effective” reproductive number. R_e depends on both the R_0 as well as the extent of immunity to the pathogen in the population ($R_e = R_0(1 - \text{fraction immune})$). Adaptation of the pathogen (for increased transmission in the human population) leads to a concomitant increase in R_0 and R_e . In contrast, waning of population immunity to the pathogen leads to an increase in R_e , but does not change R_0 (as R_0 is defined as the extent of transmission in a wholly susceptible population). We let $R_{0,Spillover}$ and $R_{0,Evolved}$ equal the basic reproductive numbers of the initial spillover and the evolved virus after it has reached a fitness peak for transmission in the human population, respectively.

In the first section we consider a basic model for evolutionary emergence of a zoonosis. In this model, there is no population level immunity against the zoonosis. In the subsequent section, we extend the basic model to consider how the probability of evolutionary emergence of a zoonosis increases as population level immunity to a related, eradicated human pathogen wanes after the cessation of vaccination.

Basic model

In Fig 1 we outline the stages for the evolutionary emergence of a novel pathogen (referred to henceforth as the zoonosis) from a zoonotic reservoir [33]. The first stage is cross-species transmission, termed spillover, that results in the introduction of infections into the human population. These spillover events can lead to chains of transmission of the pathogen in the population. If the $R_{0,Spillover} < 1$, then these chains of transmission will stutter to extinction. Evolution of the pathogen during these stuttering chains of transmission can result in an increase in R_0 and, if the resulting $R_0 > 1$, this can allow the zoonosis to persist in the human population [4, 16]. If R_0 has increased to above 1 prior to spillover then further evolution is not needed for it to persist in the human population [43]. In addition to these stages (which have been considered in earlier models) we introduce a final stage which involves the further adaptation of the zoonosis for transmission in the human population. We expect this final stage to end when the zoonosis has reached a fitness peak for transmission in the human population, after which further adaptive evolution will stop or be greatly reduced.

Probability of emergence per spillover.

We define P_E as the probability that a single spillover results in the emergence of a highly transmissible variant of the pathogen with a basic reproductive number $R_{0,Evolved}$. In calculations, we set $R_{0,Evolved} = 3$, which is well within the range

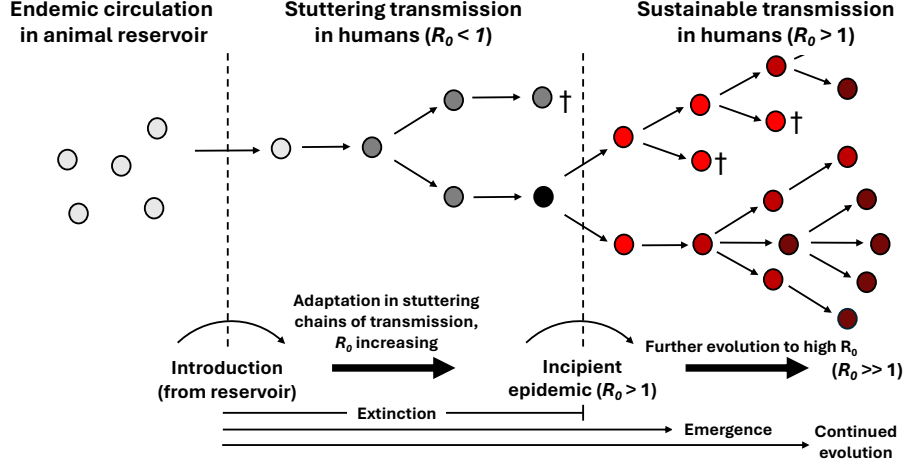


Figure 1: **Schematic showing the stages for the emergence of a pathogen into a population with cross-reactive immunity.** The pathogen in the zoonotic reservoir (circles) occasionally spills over into the human population. If R_0 in humans is less than 1 (grey circles) it can nevertheless cause chains of transmission that stutter to extinction. During these chains of transmission R_0 can increase (depicted by darker grey color) and if R_0 becomes greater than 1 (red circles) it can persist in the human population. (If $R_0 > 1$ for the spillover the above step is not needed.) Subsequent to $R_0 > 1$, adaptive evolution can increase R_0 further ending when it has reached a fitness peak in humans (lighter red circles).

of R_0 for endemic human pathogens. We use a branching process model of disease transmission (adapted from [2, 4, 21]) to consider how P_E depends on the reproductive number of the initial zoonotic infection ($R_{0,Spillover}$), the beneficial mutation rate per infection (μ) (which we subsequently call simply the mutation rate) and the fitness advantage conferred by a mutation (σ), which we assume to be a constant proportional increase in transmission. This gives us a simple additive fitness landscape with no macroscopic epistasis [17], and we discuss the effect of having other fitness landscapes in the supplement. We incorporate two additional features to the earlier models. First, as the reproductive number $R_{0,Spillover}$ of the zoonosis increases, fewer adaptive mutations are required for the reproductive number of the evolved variant to exceed 1 (see SI Fig 1 for more detail). Second, we allow adaptive evolution to continue after pathogen transmission has just become supercritical until the reproductive number of the zoonosis reaches a value comparable to that of a typical endemic disease – which we choose conservatively as $R_{0,Evolved} \approx 3$ – at which point the epidemic will be much harder to control. We refer the reader to the supplement for the branching process calculations.

In Fig 2 we plot how P_E depends on R_0 , μ , and σ . The plots illustrate how P_E increases with increasing $R_{0,Spillover}$. If $R_{0,Spillover} > 1$, P_E is relatively high, and only stochastic extinction when the numbers of infecteds is low prevents emergence. Fig 2A shows that P_E can fall very rapidly as $R_{0,Spillover}$

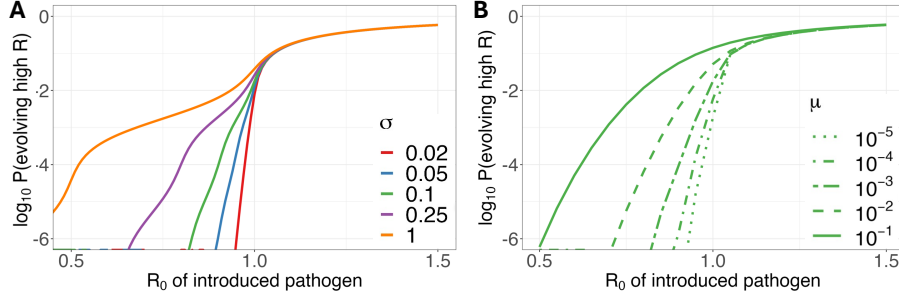


Figure 2: P_E increases with $R_{0,Spillover}$, σ , and μ . The probability of evolutionary emergence per introduction (P_E) for a pathogen depends on the reproductive number $R_{0,Spillover}$ of the zoonoses when it spills over (x-axis), the selective advantage σ of a mutation (Panel A) and the mutation rate μ (Panel B). P_E increases very steeply (the y-axis is on a log-scale) with increases in $R_{0,Spillover}$, and falls rapidly as σ and μ decrease. Parameters: $\mu = 10^{-3}$ in panel A and $\sigma = 0.1$ in panel B. The calculations for the probability of evolutionary emergence (defined as reaching an $R_0 \geq 3$) are described in the SI.

decreases below 1, and the dependence of P_E on $R_{0,Spillover}$ is stronger when the proportional increase in fitness arising from each mutation (σ) is small. This is because as σ decreases the number of mutations (n) required for R_0 to exceed one increases (as $R_{0,Spillover}$ approaches 1, $n \sim (1 - R_{0,Spillover})/\sigma$). Fig 2B shows how P_E declines as the mutation rate μ falls. When $R_{0,Spillover} < 1$ the dependence of P_E on μ is compounded by the number of mutations required in order for R_0 to exceed one as $\sim \mu^n$.

Evaluating pandemic potential (P_P).

We define the pandemic potential, P_P , as the probability of a successful emergence event (of a variant with $R_{0,Evolved} = 3$) in a year. The dependence of P_P on the probability of emergence per spillover, P_E , and the rate of spillovers which equals the average number of spillover events in a year, s , is given by

$$P_P = 1 - (1 - P_E)^s \approx sP_E \quad \text{if } sP_E \ll 1. \quad (1)$$

This is intuitive – when P_E is small, the probability of a pandemic scales linearly with the number of spillover events and saturates as the probability of spillovers causing a pandemic approaches 1. We first consider how P_P depends on the spillover rate (s) and the transmissibility of the pathogen in the human population in the absence of immunity ($R_{0,Spillover}$). We will then apply this understanding to explore how P_P for a zoonosis increases as population immunity wanes.

In Fig 3, log-fold changes in s are necessary to see the same effect on P_P as linear changes in $R_{0,Spillover}$. We therefore see that the change in the P_P is largely driven by increases in $R_{0,Spillover}$ rather than the change in s . In the SI we show that this result is robust to different μ and σ , as well as the addition

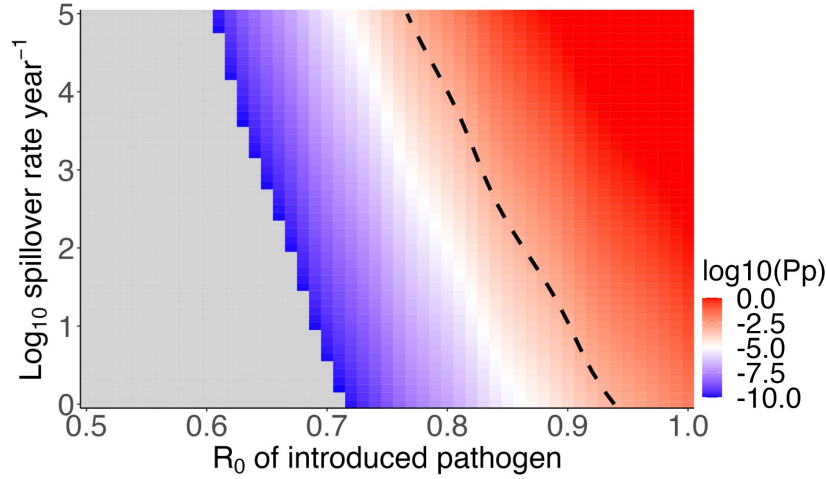


Figure 3: **Dependence of the pandemic potential P_P on R_0 of the zoonosis and the spillover rate.** We plot how P_P depends on the reproductive number of the zoonosis ($R_{0,Spillover}$) (x-axis) and the spillover rate (s) (y-axis). We assume there is no population immunity. The black dashed line demarcates the region where $P_P > 0.001$ indicating the range where evolutionary emergence is likely to pose a problem. Parameters: $\sigma = 0.1$ and $\mu = 10^{-3}$. P_P increases sharply with linear increases in $R_{0,spillover}$ but requires log-fold differences in s to change substantially.

of rare pre-adapted variants of the pathogen in the zoonotic reservoir (SI Figs 2 and 3).

The black dashed line corresponds to a time to emergence (i.e $1/P_P$) of 1000 years. If the $R_{0,Spillover}$ of the zoonotic is to the left of the black line then we would not have to worry about emergence as it is unlikely to happen in the next 1000 years. If the $R_{0,Spillover}$ of the zoonotic is to the right of the black line then to prevent emergence, we would ideally want to intervene before immunity has waned to an extent R_e has crossed to the right of the black line. We explore this in the following sections.

Consequences of ceasing vaccination after eradication

Vaccination or immunity to a human pathogen reduces both the number of spillovers of related zoonoses into the human population and the extent of human-human transmission of these zoonoses. We now consider the emergence of a zoonosis as population immunity wanes.

Waning population immunity after eradication.

Population immunity following cessation of vaccination declines due to the replacement of immune individuals when they die with newborn susceptibles – the longest that population immunity can last is the lifespan of the immune individuals. Other factors that affect the waning of immunity to the zoonoses

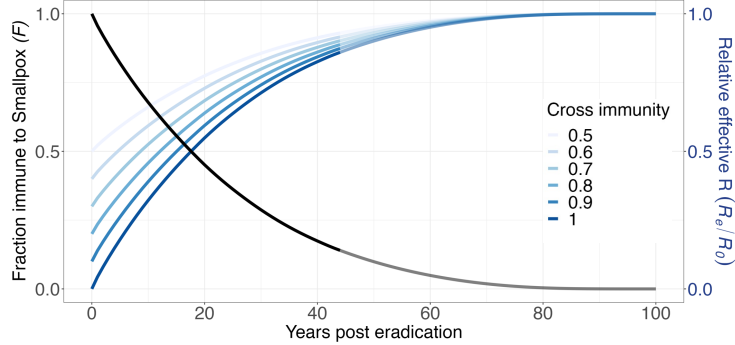


Figure 4: **We illustrate the loss of population level immunity to smallpox and the effective reproductive number scaled to its value in a naive population (i.e. R_e/R_0) for mpox** using data for demographics of the Democratic Republic of Congo and projections [42]. The left y-axis and black line indicates how the fraction of individuals in the DRC with immunity to smallpox wanes after vaccination was stopped. The right y-axis and blue lines indicate how the effective reproductive number of mpox transmission in the human population increases over time. We assume that immunity to smallpox is lifelong.

include: the waning of immunity within vaccinated individuals, limited cross-reactivity to the zoonosis, and the extent of vaccine coverage prior to cessation of vaccination. The reader is referred to Taube et al [40] for a more detailed consideration.

In Fig 4 we illustrate how population level immunity is lost for a simple scenario where all individuals are immune when vaccination ends and immunity is lifelong. The level of population immunity to the eradicated pathogen at time t after the cessation of vaccination, $F(t)$ equals

$$F(t) = \int_t^\infty f(a)da. \quad (2)$$

where $f(a)$ is the fraction of population of age a at time t after eradication and cessation of vaccination. The effective reproductive number of the zoonosis, R_e , depends both on the extent of population immunity to the eradicated pathogen, $F(t)$, and the extent to which this immunity provides cross-immunity to the zoonosis, which we describe by the parameter c :

$$R_e(t) = R_0(1 - cF(t)). \quad (3)$$

The parameters in this illustrative example pertain to smallpox, which confers cross-immunity against multiple zoonotic orthopoxviruses (for concreteness, we treat mpox as the representative virus among these initiating an epidemic). We use demographic estimates from the DRC to gain insight into the rate at which $F(t)$ might decay in populations where spillovers are frequent.

As $F(t)$ becomes small, the effect of c (and the extent of waning of immunity within individuals) on R_e diminishes and R_e approaches R_0 . If the level of

protective immunity wanes in an individual over time, then we expect R_e to increase faster than shown in these curves.

The change in population immunity to the zoonosis affects its emergence by increasing the number of spillovers in addition to the number of secondary cases arising from each spillover. The rate of spillovers (i.e., the number of spillover infections per year), $s(t)$, equals the number of individuals per year who would be infected following contact with the zoonotic reservoir. $s(t)$ depends on the rate of spillover expected in a completely susceptible population, s_0 ; the fraction of individuals with immunity to the eradicated pathogen $F(t)$; and the level of cross-protection against the zoonosis, c ; which gives us an equation for $s(t)$ nearly identical to eqn 3 reflecting that the drop in spillovers is proportional to the drop in R_e :

$$s(t) = s_0(1 - cF(t)) \quad (4)$$

Both R_e and s increase proportional to the extent to which immunity in the population wanes. Fig 3 demonstrates how P_P is significantly more sensitive to transmission than to spillover. Consequently, we focus on changes in transmissibility (R_e) to examine the effect of waning population immunity on evolutionary emergence.

Increase in pandemic potential with waning immunity.

Both the frequency of spillovers s and the R_e of spillover infections increase linearly with decreases in the extent of immunity in a population. Results shown in Fig 3 indicate changes in transmission (R_0) have a much greater impact on the emergence of a zoonosis compared to changes in the rate of spillover (s). In Fig 5A, we extend this to illustrate how pandemic potential increases as population immunity wanes over times – i.e. as R_e and s increase over time. The lines indicate possible trajectories of P_P over time post eradication (black labels), according to the basic reproductive numbers $R_{0,Spillover}$ of the spillover virus. We use our estimates of the loss of population immunity over time from Fig 4 with cross-protection $c = 0.8$ (chosen to be in the ballpark of estimates for mpox [12, 15, 20]), and let the spillover rate $s_0 = 10^4$. As population immunity wanes over time, R_e and s increase, and the color of the heatmap indicates the corresponding increase in pandemic potential. We see that the increase in the pandemic potential over time indeed occurs due to changes in R_e (going to the right) rather than increases in the spillover rate (going vertically).

While the trajectories are to the left of the dashed line, the pandemic potential is less than 1 in 1000 per year and so emergence is not of concern. When P_P becomes greater than 10^{-3} , this indicates pandemic emergence is possible and increasingly likely. In Fig 5B we plot how the pandemic potential increases for the trajectories shown in Fig 5A.

To the left of the vertical dotted line Fig 5B, pandemic emergence is not just unlikely, but impossible, because population immunity is sufficient to keep R_e of the fully evolved variant below 1 (i.e $R_{0,Evolved}(1 - cF) < 1$). If $R_{0,spillover}$ is sufficiently low, the pandemic potential never increases above $P_P = 10^{-3}$

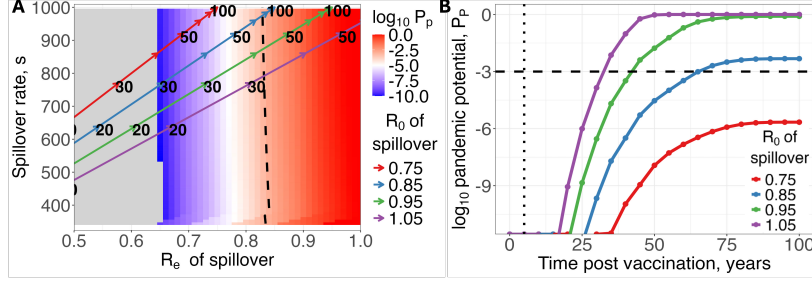


Figure 5: **Pandemic potential P_P increases largely due to R_e .** (Panel A) We plot how P_P (probability of emergence per year) depends on R_e (x-axis) and s (y-axis). The dashed line demarcates the region where $P_P < 0.001$ indicating the range where evolutionary emergence is likely to pose a problem. The colored lines with arrows show trajectories of s and R_e due to waning immunity for different R_0 . Black labels show years post eradication. We see the waning of population immunity (taken from Fig 4) can result in a large increase in P_P , and this occurs largely due to the increase in R_e rather than s . (Panel B) We show how the pandemic potential P_P increases over time and its trajectory depends on $R_{0,Spillover}$. Parameters: $s_0 = 10^4$, $\sigma = 0.1$ and $\mu = 10^{-3}$.

(horizontal dashed line), indicating a pandemic is not of concern on a relevant timescale. If $R_{0,Spillover}$ is sufficiently close to 1, emergence will have become likely by the time population immunity has waned completely. The time at which P_P crosses the dashed line indicates the amount of time after ceasing vaccination before pandemic emergence becomes a concern.

Interventions to prevent emergence

Interventions to reduce the pandemic potential, P_P , can do so by reducing R_e (the transmissibility of the pathogen) or reducing s (the rate of spillovers)—the parameters μ and σ are properties of the pathogen-host system and are not under human control. We now consider the effect of interventions on reducing P_P .

Some interventions can reduce P_P by reducing R_e alone. This could be accomplished by building accessible and efficient public-health and clinical health-care systems. These could incorporate rapid testing, a treatment reducing the duration of infection, as well as quarantine measures as applicable.

Other interventions can reduce P_P by reducing the number of spillovers, s , alone. This is accomplished by reducing the ecological overlap between the zoonotic reservoir and human hosts [33].

Our simulations suggest this intervention would likely to result in only a modest reduction in P_P , compared with the much larger reductions that would be obtained by interventions such as vaccination that decrease R_e . However, this result does not take into account the costs to achieve similar fold reductions in these parameters, the potential for the reductions to cover multiple pathogens, or other beneficial effects of reducing s .

Immunization can reduce P_P by decreasing both s and R_e . As described

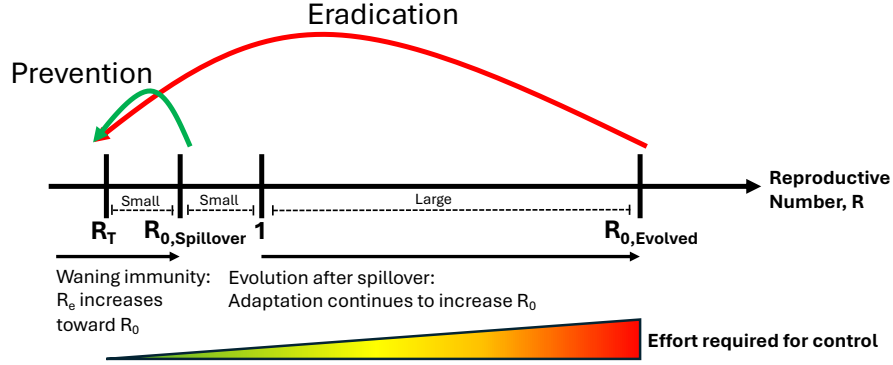


Figure 6: **Eradication requires a greater intervention than prevention.** The magnitude of the response required to control disease after its evolution to high transmissibility ($R_{0,Evolved}$) equals $(1 - 1/R_{0,Evolved})$. This is significantly greater than the magnitude required to prevent the emergence of the disease when its reproductive number is still small (near $R_{0,Spillover}$), which equals $(1 - R_T/R_{0,Spillover})$. R_T is the R_e for which P_P is sufficiently low.

earlier, the decrease in R_e will have a much greater contribution to reducing P_P than the decrease in s .

Costs of delaying intervention

It is widely held that prevention is better than a cure, and this is true for the emergence of a zoonosis as illustrated in Fig 6. In its simplest form, prevention might occur by ensuring R_e (or more accurately combination of R_e and s) is kept below a critical level (R_T) that corresponds to an acceptably low probability of evolutionary emergence as depicted in Fig 6. We see that, for a wide range of parameters, this requires keeping R_e to a level slightly below 1 (the exact level, R_T , depends on σ and μ). This might entail, for example, a 20% reduction in transmission (if, for example, it requires reducing R_0 of the zoonotic infection from 1 to 0.8). In contrast, if the zoonosis has had time to evolve its transmissibility to a level commensurate with an endemic pathogen (which have set to a conservative level of $R_{0,evolved} = 3$ in our simulations), then pathogen elimination requires preventing $1 - 1/R_0 = 2/3$ or 66% of all transmissions. Consequently, a much lower level of intervention such as vaccination is needed to prevent a pandemic rather than control it after it has occurred and the has adapted to transmit in the human population as illustrated in Fig 6.

Discussion

Prior to the eradication of an endemic human pathogen, infections with related zoonoses are not a problem. This is because cross-immunity arising from infection or vaccination against the endemic pathogen reduces both zoonotic spillovers as well as human-human transmission of the zoonoses. This is an example of the principle of competitive exclusion operating on pathogens [8]. Eradication and cessation of vaccination leads to the waning of immunity to the related zoonoses and opens up a niche for related zoonoses [35]. This leads to an increase in both the number of zoonotic cases and the extent of human-human transmission, which potentiate further adaptation leading to evolutionary emergence of the zoonosis into the human population. We have used mathematical models to explore how waning immunity can lead to evolutionary emergence of a highly transmissible human pathogen.

We extended a previous model [4] of the evolution of a pathogen during stuttering chains of transmission after spillover to consider the emergence in three ways. First, we incorporated changes (an increase) in the effective reproductive number (R_e) for transmission of the zoonotic pathogen between humans as cross-immunity waned. Second, we linked the increases in R_e to a decreasing number of mutations needed for the transmission to become supercritical. Finally, we allowed evolution to continue even after the R_e of the zoonosis in the human population is greater than 1 (i.e. until its basic reproductive number $R_{0,Evolved} = 3$ in our simulations)

Our analysis showed that, as R_e approaches 1, there is a very rapid rise in the probability of evolutionary emergence, and the dependence of P_P on R_e is sharper than previously estimated in [4] due to the decline in the number of mutations required as R_e approaches 1. We find that for a zoonosis to be a concern requires its $R_{0,Spillover}$ to be sufficiently high (to the right of the black line in Fig 3A). As population immunity wanes we find that P_P can change dramatically. Because the waning of population immunity occurs with a timescale of a human lifespan (or shorter if it wanes over time in an individual), we observe that the zoonoses can shift from being of minor concern to a major problem in the course of a decade or so as illustrated in Fig 5.

Interventions to prevent the evolutionary emergence of a zoonosis can do so by reducing the frequency of spillover infections (s) or the extent of transmission in the human population (R_e). Our results show that the latter (decreasing R_e) is much more effective than the former (decreasing s). Ecological interventions that reduce contact between the zoonotic reservoir and humans will proportionally reduce the number of zoonotic infections but unless the number of spillover events is very low, our results suggest that this is unlikely to have a substantial effect on the probability of evolutionary emergence. Other interventions such as improvements in surveillance and universal access to healthcare reduce R_e and this can lead to a much greater reduction in the probability of evolutionary emergence, particularly in the regime where $R_0 \sim 1$. These interventions also benefit many aspects of the health of communities.

Vaccination can prevent evolutionary emergence by reducing both the spillover

rate and transmission of a disease, and our study suggests that its effect is largely from the extent of reduction in transmission. Vaccines can work in multiple ways: they can reduce the susceptibility of individuals to infection and, in those who do get infected, they can reduce the infectivity and the extent of pathology of the infection [19]. Preventing the evolutionary emergence of a zoonosis by vaccination would benefit from vaccines that are designed to confer broad cross-reactive immunity even if this is at the expense of generating immunity to the original human pathogen. It would also require a focus on reducing transmission (by a combination of reducing susceptibility and infectivity) rather than pathology. Indeed, a vaccine that only reduces pathology might increase the probability of emergence due to transmission from asymptomatic infections. It may even be that permitting some symptoms can allow for the better detection of spillover infections and trigger measures that reduce transmission [14].

Our most salient result is that the level of vaccination (or other intervention) required to prevent evolutionary emergence is modest compared to that needed for control and eradication after it has adapted to become highly transmissible human pathogen. Small reductions in transmission greatly reduce the evolutionary potential of the zoonosis when $R_{0,Spillover} \sim 1$ (even if slightly supercritical R_0), but after $R_{0,Evolved} \gg 1$ eradication requires a vaccination of a fraction ($f = 1 - 1/R_{0,Evolved}$) of the population. Due to the persistence of the pathogen in zoonotic reservoirs, however, this low-level of vaccination (or other intervention) needs to be ongoing and introduces an interesting difficulty with the counterfactual. To what extent can we know the true necessity of ongoing prevention efforts if the disease never emerges because of them? We leave this question to the reader as a philosophical exercise.

The epidemiology of mpox indicates limitations of our understanding

The eradication of smallpox by vaccination is considered one of the crowning achievements of modern medicine. Since smallpox did not have a zoonotic reservoir and since mpox did not pose a significant problem (at least at that point in time) eradication was followed by the cessation of vaccination in 1984. The cessation of vaccination was motivated by the costs of the vaccination program and potential, though rare, adverse events associated with vaccination. Over time the resultant reduction in population immunity to smallpox and cross-immunity to mpox has resulted in an increase in both the number of spillover events as well as the spread of the virus within the human population. It was initially thought that the R_0 of mpox was slightly below one and consequently its emergence would not be a problem [12]. However subsequent studies suggested that when R_0 was only slightly below one adaptation of mpox would likely lead to evolutionary emergence (i.e. mpox variants having $R_0 > 1$) allowing for sustained transmission in the human population [4, 27, 35, 36]. Our study refines the predictions of these earlier papers and their consequences for intervention.

Recent observations and studies highlight concern about the potential of evolutionary emergence of mpox. The fraction of individuals with smallpox immunity has declined such that R_e has increased from well below one in the 1980's (estimated at around 0.3 [7]) to close to or likely above one in the next decade when the population has almost no immunity to smallpox. The increasing frequency and intensity of mpox outbreaks over this time frame are consistent with the aforementioned models. In recent years, the R_e of mpox has reached around 1 in groups where close contact, typically during sex, facilitates its transmission. This has resulted in the disease persisting in these groups, even in the absence of ongoing zoonotic introductions [13]. Fortunately, the reproductive number of both clade 1 and 2 strains in these groups is only slightly greater than one, placing it in the regime in which modest public health interventions and targeted vaccination may prevent continued evolution but may not eliminate these outbreaks. Over time, the waning of residual population immunity may allow a further increase in the R_e of mpox facilitating sustained transmission in a larger fraction of the population. This remains to be seen.

The ongoing human to human transmission in the high risk groups in which it is currently endemic raises the probability of it adapting to increase transmission between humans in this and the general population. This would result in mpox being able to spread in more communities, and necessitate much more widespread intervention for its control. We now discuss some of the complexities associated with mpox epidemiology and evolution that limit our ability to make quantitative predictions of its evolutionary trajectory.

Mpox could adapt by (i) increasing its ability to replicate within individuals, (ii) evading the immune response, or (iii) transmitting via a respiratory route in addition to the very close contact needed for transmission of the currently circulating strains [9, 18]. A major limitation is our lack of knowledge of the mutational and adaptive landscapes that underlie these changes, and re-

search in this area is necessarily limited by the safety of experimental evolution of mammalian-infecting viruses [26]. It is likely that virus adaptation requires multiple mutations, and this possibly contributes to why we have not yet seen variants with much higher fitness in clades 1 and 2 despite their having infected over 50,000 and 100,000 individuals respectively since their most recent emergence [13]. The need for multiple mutations for adaptive evolution was noted for SARS-CoV-2, where it took several months before the more transmissible delta and omicron variants emerged – and these viruses had dozens of mutations, which likely arose in immunocompromised individuals [37, 41]. Circulating mpox variants contain many more mutations than expected, but this is most likely due to editing by human APOBEC3 enzyme, which are indicative of sustained human to human transmission as opposed to multiple zoonotic introductions or adaptation [32, 39]. Whether some of these mutations do increase, or further mutations (potentially emerging from sustained infections in immunocompromised individuals) will increase, mpox fitness in the human host remains to be seen. We also note that our branching process calculations estimate the probability that a spillover will give rise to a highly transmissible variant at some point in the future, but not how long this process will take, nor how changes the prevalence of infections during this time may trigger intervention [23]. The time between the spillover event and the generation of the evolved variant will depend on the length of the intervening chain of transmission as well as the serial interval (which is relatively long, ~ 2 weeks [29, 34].)

This paper focuses on how the waning of cross-protective population immunity facilitates the emergence of a zoonosis and does so for the simple case of a well-mixed population. The next steps would be to consider heterogeneity in transmission events, which has been the subject of considerable interest [6, 11, 28, 44]. For the case of mpox, this would involve consideration of its transmission in highly connected networks (i.e sex workers and MSM). The transmission of mpox in these groups may be different from that of HIV and other STDs that result in persistent infections. The persistent nature of HIV allows a highly connected infected individual to spread the infection for an extended duration. In contrast, acute infections such as mpox are of shorter duration, and, following recovery, surviving individuals are immune, which could potentially limit further spread. Regardless, as population level immunity is lost we would expect to see the first signs of increased transmission in outbreaks among high transmission groups. Another problem common to emerging pathogens in general is that the spread of the pathogen can result in changes in the contact patterns of humans and modify its spread, and the extent of these changes can vary dramatically between different locations. This was seen following the initial emergence of SARS-CoV-2, where some countries could prevent successful introductions (e.g., New Zealand), a few could introduce sufficiently severe lockdowns that resulted in local eradication (e.g., China) and most could at best flatten or reduce the number of cases prior to development of a vaccine [24, 25].

In conclusion, our limited understanding of disease emergence arises both from complexities in the evolutionary and epidemiological processes involved and the paucity of examples of pathogens that have successfully emerged into the

human population recently enough to allow detailed analysis. Indeed, mpox is currently the strongest example of a zoonotic disease for which the loss of cross-protective herd immunity might facilitate its emergence. Still, while the actual eradication of smallpox required a far more nuanced strategy that included surveillance followed by ring vaccination, the simple formulation of a critical threshold for effective vaccination ($f > 1 - 1/R_0$) provided the rationale for the eradication of smallpox. Similarly, the simple analysis we describe highlights the potential risk of evolutionary emergence associated with disease eradication and the subsequent waning of immunity to related zoonoses. The considerations outlined in this paper might inform us of how early and ongoing intervention might be needed following the eradication of pathogens such as smallpox.

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